

CVaR-Based Portfolio Optimisation for a Malta-Accessible UCITS ETF Basket

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Abstract—This paper examines CVaR-based portfolio optimisation for a Malta-accessible UCITS ETF basket, focusing on whether a simple downside-risk optimisation framework can improve risk control compared with Equal-Weight and Minimum-Variance benchmarks. The aim is to evaluate the three strategies under identical long-only and fully invested conditions using historical ETF market-price data converted to EUR. The proposed solution implements a Python-based rolling out-of-sample backtest with a 252-trading-day estimation window, 21-trading-day rebalancing, and performance evaluation using CAGR, volatility, Sharpe ratio, Sortino ratio, VaR, CVaR, and maximum drawdown. The results show that CVaR-95 achieved the lowest VaR and CVaR, but Equal-Weight delivered the strongest CAGR and Sharpe ratio, while Minimum-Variance achieved the lowest maximum drawdown; therefore, the hypothesis is only partially supported. The project was implemented using Python, Jupyter Notebook, pandas, NumPy, SciPy, CVXPY, and yfinance-based historical data, with rolling-window backtesting, long-only optimisation, and CVaR sensitivity analysis as the main techniques.

Index Terms—CVaR, portfolio optimisation, UCITS ETFs, Equal-Weight, Minimum-Variance, rolling backtesting, Python.

I. INTRODUCTION

Recent market crises, including the COVID-19 shock, have renewed criticism of classical portfolio theory. Although the mean-variance (MV) framework remains foundational, variance treats upside and downside deviations symmetrically and therefore does not isolate severe losses in the way downside-focused investors typically require [1], [2]. For this reason, portfolio research has increasingly adopted Conditional Value-at-Risk (CVaR), which estimates average losses beyond the VaR threshold and can be incorporated into tractable optimisation formulations [2], [3]. Exchange-traded funds (ETFs) provide a practical setting for such comparison because prior studies have evaluated CVaR-based, MV, and equal-weight approaches using diversified ETF universes and out-of-sample backtesting [3].

Despite this relevance, much of the recent CVaR literature is embedded in advanced settings such as evolutionary algorithms, copula-based dependence modelling, bilevel multi-market optimisation, deep reinforcement learning, and distributionally robust optimisation [2]–[6]. These studies are valuable, but they make it harder to judge whether a simpler

and more transparent CVaR-minimisation framework, implemented under standard long-only constraints, already provides meaningful downside-risk benefits. In addition, while Malta has an established collective investment scheme environment and the Maltese Assets Fund illustrates the local relevance of diversified fund products [7], [8], there is limited evidence on CVaR-based optimisation in a Malta-accessible UCITS ETF context. The present study is therefore positioned at the intersection of quantitative portfolio theory and applied investment management, addressing a contextual and empirical gap by testing whether a Malta-accessible UCITS ETF basket can serve as a credible benchmark setting for comparing CVaR, Equal-Weight, and Minimum-Variance strategies.

The purpose of this quantitative backtesting study is to evaluate whether CVaR-minimisation provides stronger downside-risk control than Equal-Weight (1/N) and Minimum-Variance portfolio construction in a Malta-accessible UCITS ETF basket over 2016–2025. The independent variable is the optimisation strategy, operationalised through three portfolio rules: Equal-Weight, Minimum-Variance, and CVaR-minimisation. The dependent variables are the observed portfolio outcomes, namely CAGR, volatility, Sharpe ratio, Sortino ratio, VaR, CVaR, and maximum drawdown. To preserve comparability, the ETF universe, estimation window length, rebalancing frequency, and long-only fully invested constraints are treated as control variables. The study aims to implement the three strategies under identical conditions using rolling out-of-sample backtesting and to compare their performance on both return-based and downside-risk criteria.

Accordingly, the main hypothesis states that a CVaR-minimised portfolio will achieve lower CVaR and lower maximum drawdown than Equal-Weight and Minimum-Variance portfolios without material deterioration in CAGR or Sharpe ratio. The null hypothesis states that no meaningful difference will exist among the three strategies in CVaR, maximum drawdown, or risk-adjusted performance. As shown in Fig. 1, the present study adopts a positivist philosophy and a deductive approach. It employs a controlled experimental backtesting strategy using secondary historical ETF data, a mono-method quantitative design, and a longitudinal time horizon. The techniques and procedures consist of secondary ETF data collection, rolling out-of-sample backtesting, and quantitative

performance and downside-risk analysis. Specifically, portfolio outcomes are treated as objectively measurable through predefined financial metrics. A deductive approach is adopted because the study begins from established theory on the limitations of variance as a risk proxy and tests whether a downside-risk measure produces superior empirical outcomes [1], [2]. Since performance is evaluated across repeated windows over 2016–2025 rather than at a single point in time, the time horizon is longitudinal.

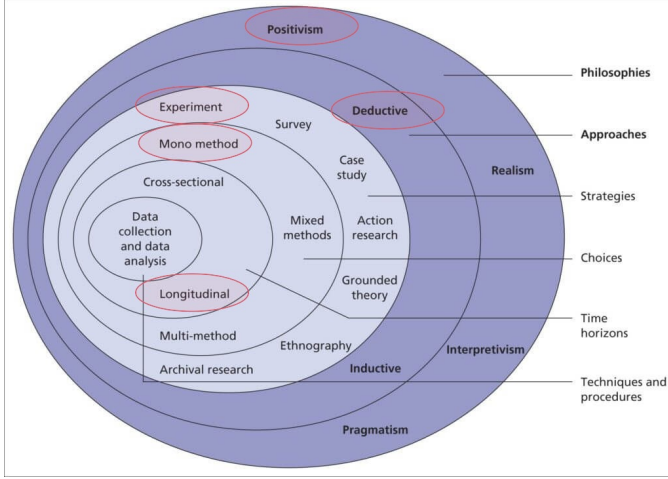


Fig. 1. Research onion for the present study, adapted from Fig. 4.1 in Saunders, Lewis, and Thornhill [9].

II. LITERATURE REVIEW

Section II reviews the research methodologies used in six peer-reviewed journal articles [1]– [6]. Non-academic and institutional sources are used only for contextual background and are excluded from the methodological comparison.

Fig. 2 organises the reviewed methodological literature into data employed, process and candidate techniques, and evaluation and testing strategies. The map shows that prior studies mainly rely on secondary financial data and backtesting-based evaluation, but differ substantially in portfolio-construction complexity. The final branch highlights the gap addressed by the present study, namely a simpler and more transparent comparison of Equal-Weight, Minimum-Variance, and CVaR-minimisation strategies in a Malta-accessible UCITS ETF setting under identical long-only conditions.

A. Overview of reviewed methodologies

A consistent methodological pattern across the reviewed studies is the use of a positivist, quantitative design based on secondary financial data. Rather than using interviews or surveys, the reviewed papers typically rely on historical price or return series and assess portfolio-construction rules through computational testing. This broad methodological direction aligns closely with the present study, which also examines objectively measurable portfolio outcomes under controlled backtesting conditions. Accordingly, the literature map for the

present study is organised around data, process, and evaluation, as illustrated in Fig. 2.

B. Data employed in prior studies

The reviewed literature shows that portfolio-optimisation research commonly uses historical market data covering several years so that both normal and stressed market conditions are represented. Muñoz and Cifuentes [1], for example, analyse historical returns from 2004 to 2023 and deliberately include major crisis periods to test the stability of mean-variance and CVaR formulations. Their use of rolling windows is methodologically valuable because it allows repeated comparison across time rather than relying on a single in-sample period.

Teplova et al. [3] also use a long historical dataset, drawing on daily returns from international ETFs between 2012 and 2022. This is particularly relevant because ETFs are also the asset class used in the present study. Their data design supports realistic comparison between portfolio rules while also enabling out-of-sample testing.

By contrast, Nasini et al. [4] use a much larger firm-level dataset across thousands of U.S. listed companies, while Cui et al. [5] focus on cryptocurrency returns. Kobayashi et al. [6] also work with large asset universes in a robust optimisation setting. These studies show that broader datasets are possible, but they also increase modelling and computational complexity. For the current research, a Malta-accessible UCITS ETF basket is more appropriate because it remains sufficiently diversified while preserving transparency, feasibility, and direct relevance to the stated research context.

Overall, the literature suggests that the most suitable data strategy for the present study is a secondary dataset of historical ETF prices and returns, covering a sufficiently long period to include both calm and turbulent market conditions, while keeping the investable universe constant across all tested strategies.

C. Portfolio-construction process and candidate techniques

The reviewed studies show that the main methodological distinction lies in the portfolio-construction process. Some studies compare classical benchmark approaches, while others embed CVaR within more advanced optimisation frameworks.

Muñoz and Cifuentes [1] provide the clearest benchmark-comparison methodology. They compare mean-variance and CVaR formulations under the same target-return structure, which isolates the effect of the risk metric itself. This is highly relevant to the present study because it demonstrates how a fair experimental comparison can be achieved when constraints and data inputs are held constant.

Teplova et al. [3] extend the process stage through copula modelling and a Black-Litterman mean-CVaR framework. Their methodology is rigorous, but it depends on parameter calibration, investor-view modelling, and dependence-structure estimation. Kaucic et al. [2] instead treat the process stage as a multi-objective evolutionary search problem, comparing variants of NSGA-II and SPEA 2 under downside-risk measures such as semi-variance and CVaR. Nasini et al. [4] introduce

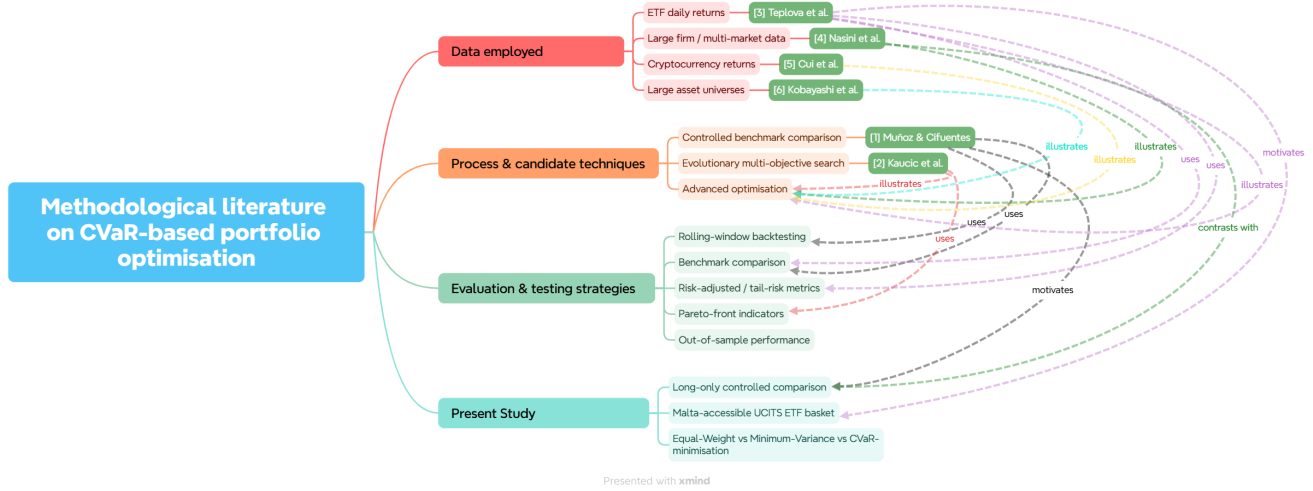


Fig. 2. Methodological literature map for CVaR-based portfolio optimisation.

a bilevel multi-market formulation, while Kobayashi et al. [6] use distributionally robust optimisation with cardinality constraints. Cui et al. [5] move further still by integrating deep reinforcement learning into a mean-CVaR framework.

These advanced methodologies are useful as evidence that CVaR is a mature and flexible risk measure, but they are not ideal templates for the present project. The current study is not intended to propose a highly specialised optimisation engine. Instead, it aims to compare three portfolio-construction rules under identical conditions: Equal-Weight, Minimum-Variance, and CVaR-minimisation. This choice is methodologically stronger for the present study because it prioritises interpretability, control, and reproducibility over algorithmic novelty.

D. Evaluation and testing strategies

The literature also shows a clear methodological pattern in the evaluation stage. Studies rarely stop at in-sample optimisation. Instead, they use out-of-sample testing and compare strategies through financial performance and risk metrics.

Muñoz and Cifuentes [1] use rolling windows and assess the similarity or difference between resulting portfolios under different risk formulations. Teplova et al. [3] evaluate portfolio performance against benchmark strategies and report risk-adjusted outcomes as well as tail-risk behaviour. Kobayashi et al. [6] also use out-of-sample investment performance measures, while Kaucic et al. [2] assess optimisation quality through Pareto-front indicators in addition to portfolio risk measures.

Across the reviewed studies, the most relevant evaluation measures for the present research are those that capture both return and downside risk. In particular, CAGR, volatility, Sharpe ratio, Sortino ratio, VaR, CVaR, and maximum draw-down appear suitable because they allow direct comparison between standard performance and tail-risk protection. This

is important because the study hypothesis does not claim that CVaR-minimisation will maximise returns; rather, it tests whether CVaR-minimisation improves downside-risk control without materially worsening return-based outcomes.

E. Comparative synthesis and methodological gap

Taken together, the reviewed literature supports three main conclusions. First, the dominant research tradition in this area is quantitative, secondary-data, and backtesting-based. This strongly supports the philosophical and methodological position already established in Section I and illustrated in Fig. 1.

Second, the strongest comparative designs are those that keep the dataset, constraints, and testing framework constant while varying only the portfolio rule. This is the methodological logic most clearly visible in Muñoz and Cifuentes [1], and it is the one most appropriate for the present study.

Third, although advanced methods such as copulas, bilevel programming, evolutionary optimisation, robust optimisation, and deep reinforcement learning increase technical novelty, they also introduce greater complexity and reduce transparency. For this research a simpler controlled comparison of benchmark strategies is methodologically more appropriate.

Finally, across the reviewed studies, the main independent element is the portfolio-construction or optimisation method, while the dependent outcomes are portfolio performance and downside-risk measures assessed through out-of-sample testing. Common testing strategies include rolling-window backtesting, benchmark comparison, and risk-adjusted performance evaluation. However, despite this methodological maturity, limited evidence compares CVaR-minimisation, minimum-variance, and equal-weight strategies in a simple, transparent, Malta-accessible UCITS ETF setting under identical long-only backtesting conditions.

III. RESEARCH METHODOLOGY

Portfolio construction depends heavily on how risk is defined and measured. Although mean-variance optimisation remains a classical benchmark, variance treats upside and downside deviations symmetrically and may therefore understate the practical importance of severe losses during turbulent market conditions [1], [3]. By contrast, Conditional Value-at-Risk (CVaR) focuses on the average loss beyond the VaR threshold and has been widely adopted in portfolio research as a downside-risk measure that is more sensitive to tail losses [1]–[3]. However, much of the recent literature applies CVaR within complex optimisation settings such as copula-based models, evolutionary algorithms, robust optimisation, or deep reinforcement learning [2]–[6]. This makes it less clear whether a simpler and more transparent CVaR-minimisation framework can already provide meaningful downside-risk benefits when compared with standard benchmark strategies.

In view of this problem, the present study examined whether portfolio construction based on CVaR-minimisation offers stronger downside-risk protection than Equal-Weight and Minimum-Variance strategies in a Malta-accessible UCITS ETF basket. The comparison is conducted under identical long-only and fully invested conditions using secondary historical ETF data and rolling out-of-sample backtesting. This design allows the effect of the portfolio-construction rule to be examined more directly, while keeping the dataset, constraints, and testing framework consistent across all strategies [5], [6].

The main hypothesis of this study states that, when applied to the same Malta-accessible UCITS ETF basket under identical rolling out-of-sample backtesting conditions, a CVaR-minimised portfolio will achieve lower CVaR and lower maximum drawdown than Equal-Weight and Minimum-Variance portfolios, without material deterioration in CAGR or Sharpe ratio. The corresponding null hypothesis states that no meaningful difference will exist among the three strategies in terms of CVaR, maximum drawdown, or risk-adjusted performance. This hypothesis is testable through empirical investigation because the independent variable is the portfolio-construction rule, while the dependent variables are the observed portfolio outcomes measured through predefined financial metrics [1], [5].

In light of the above, the aim of this research is to assess whether a simpler and more transparent CVaR-minimisation framework can provide stronger downside-risk protection than Equal-Weight and Minimum-Variance portfolio construction in a Malta-accessible UCITS ETF basket. More specifically, the study seeks to implement the three strategies under identical long-only and fully invested conditions and to evaluate their out-of-sample performance over the 2016–2025 period. The expected outcome is a clear empirical comparison showing whether CVaR-minimisation improves downside-risk measures such as CVaR and maximum drawdown, while preserving broadly competitive return and risk-adjusted performance.

The research objectives were as follows:

- 1) Construct a consistent historical dataset for the selected

Malta-accessible UCITS ETF basket and prepare it for portfolio backtesting.

- 2) Implement Equal-Weight, Minimum-Variance, and CVaR-minimisation portfolio-construction rules under identical investment constraints.
- 3) Conduct rolling out-of-sample backtesting to evaluate how the three strategies perform across different market conditions over 2016–2025.
- 4) Compare the strategies using return, volatility, risk-adjusted, and downside-risk measures, namely CAGR, volatility, Sharpe ratio, Sortino ratio, VaR, CVaR, and maximum drawdown.
- 5) Assess whether CVaR-minimisation provides materially stronger downside-risk control than the benchmark strategies without significant deterioration in return-based performance.

The research questions addressed in this study are organised around the data, portfolio-construction process, and evaluation stages identified in Section II. Since the present research does not propose a new complex optimisation engine, the questions are designed to examine whether a transparent benchmark comparison based on secondary ETF data can produce meaningful evidence on downside-risk control.

The research questions are as follows:

- 1) Historical dataset: What historical ETF dataset and investable UCITS ETF basket are most suitable for a controlled comparison of Equal-Weight, Minimum-Variance, and CVaR-minimisation strategies in a Malta-accessible setting?
- 2) process / candidate techniques: Under identical long-only and fully invested constraints, how does CVaR-minimisation compare with Equal-Weight and Minimum-Variance as a portfolio-construction approach?
- 3) evaluation and testing: How does the proposed CVaR-minimisation strategy compare with the benchmark strategies in rolling out-of-sample backtesting when evaluated using CAGR, volatility, Sharpe ratio, Sortino ratio, VaR, CVaR, and maximum drawdown?

A. Research Pipeline

To answer the above research questions, the study was structured around the research pipeline illustrated in Fig. 3. The first stage consisted of defining the six-ETF UCITS basket and collecting the historical adjusted closing prices required for the backtest. The second stage involved aligning prices across common trading dates, checking completeness, handling missing observations, and deriving periodic returns from the cleaned series. The third stage consisted of constructing Equal-Weight, Minimum-Variance, and CVaR-minimisation portfolios under identical long-only and fully invested constraints. The fourth stage involved rolling out-of-sample backtesting, where portfolio weights were estimated from the preceding in-sample window and then applied to the subsequent out-of-sample period until the end of 2025. The final stage consisted of evaluating realised portfolio outcomes

using CAGR, annualised volatility, Sharpe ratio, Sortino ratio, VaR, CVaR, and maximum drawdown, and comparing the CVaR strategy against the benchmark portfolios.

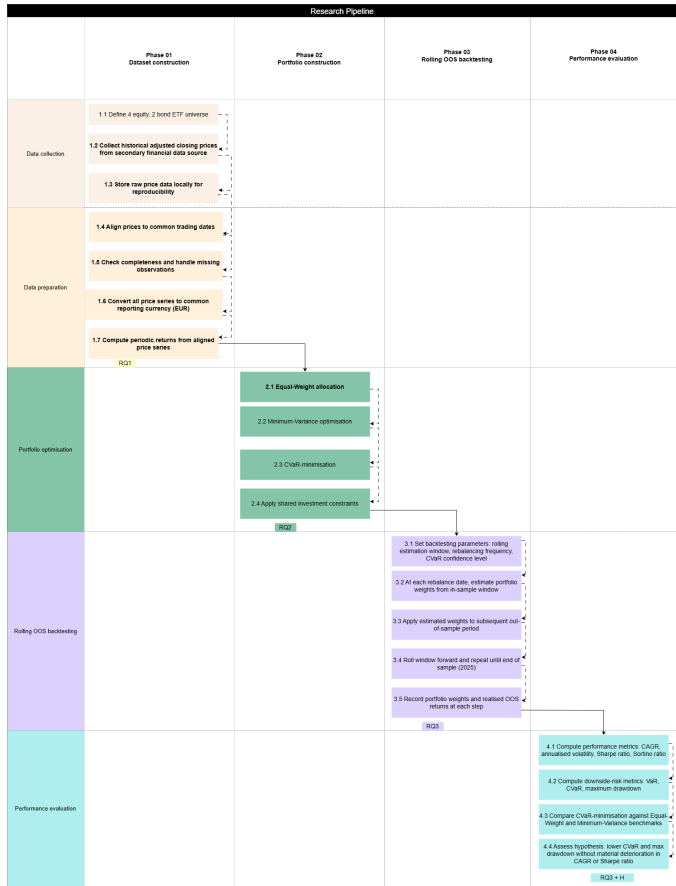


Fig. 3. Fig. 3. Research pipeline for CVaR-based portfolio optimisation in a Malta-accessible UCITS ETF basket.

B. Research Approach and Experimental Variables

Variable type	Variable	Operationalisation
Independent variable	Portfolio-construction strategy	Equal-Weight, Minimum-Variance, CVaR-minimisation.
Dependent variables	Portfolio outcomes	CAGR, annualised volatility, Sharpe ratio, Sortino ratio, VaR, CVaR, maximum draw-down.
Control variables	ETF universe	Fixed six-ETF Malta-accessible UCITS basket.
Control variables	Data treatment	Adjusted closing prices, common trading dates, identical return-construction procedure.
Control variables	Investment constraints	Long-only, fully invested.
Control variables	Backtesting settings	Same rolling estimation logic, rebalancing structure, and evaluation period for all strategies.

The present study adopted a positivist philosophy, a deductive approach, and a mono-method quantitative design. This choice was appropriate because the research aimed to test whether changing the portfolio-construction rule produced measurable differences in objectively observed portfolio outcomes. More specifically, the study employed a controlled experimental backtesting strategy using secondary historical ETF data. This design was methodologically suitable because the ETF universe, data treatment, and investment constraints were held constant, while the portfolio-construction rule was varied across Equal-Weight, Minimum-Variance, and CVaR-minimisation. In this way, the study was designed to isolate the effect of the optimisation approach as clearly as possible.

The independent variable in the study was the portfolio-construction strategy, operationalised through three portfolio rules: Equal-Weight, Minimum-Variance, and CVaR-minimisation. The dependent variables were the observed portfolio outcomes, namely CAGR, annualised volatility, Sharpe ratio, Sortino ratio, VaR, CVaR, and maximum drawdown. To preserve comparability, the ETF universe, adjusted-price treatment, common-date alignment, long-only fully invested constraints, and the same rolling out-of-sample testing framework were treated as control variables. Since the study evaluated repeated portfolio outcomes over time rather than collecting subjective human responses, the design was consistent with a quantitative backtesting methodology.

C. Dataset and Data Preparation

The dataset used in this study consisted of secondary historical ETF market-price data for a fixed basket of six Malta-accessible UCITS ETFs, representing four equity sleeves and two government-bond sleeves. The selected basket comprised iShares Core S&P 500 UCITS ETF USD (Acc), iShares Core MSCI Europe UCITS ETF EUR (Acc), iShares Core MSCI Japan IMI UCITS ETF, iShares Core MSCI Emerging Markets IMI UCITS ETF (Acc), iShares USD Treasury Bond

7–10yr UCITS ETF (Acc), and iShares Euro Government Bond 7–10yr UCITS ETF (Acc). ISINs were treated as the master identifiers for the basket, while EUR was used as the reporting currency for analysis.

Historical adjusted closing prices were downloaded as an exchange market-price dataset through the yfinance interface and stored locally for reproducibility. Adjusted prices were used because they provided a more consistent basis for return calculation than unadjusted closing prices. The raw data were then aligned across common trading dates, checked for completeness, and cleaned to handle missing observations before periodic returns were computed from the aligned series. This preparation stage was necessary to ensure that all three portfolio-construction rules were tested on the same underlying data under identical conditions.

D. Portfolio-Construction and Rolling Out-of-Sample Backtesting

Three portfolio-construction strategies were implemented in the study. Equal-Weight served as a transparent benchmark in which capital was allocated equally across all ETFs. Minimum-Variance represented the classical variance-based optimisation approach, while CVaR-minimisation represented the downside-risk approach under investigation. All three strategies were implemented under shared investment constraints so that no short selling was permitted and the full portfolio weight remained invested at each rebalance point. This ensured that the comparison remained controlled, interpretable, and directly aligned with the hypothesis.

To evaluate these strategies, the study employed rolling out-of-sample backtesting over the 2016–2025 period. At each rebalance date, portfolio weights were estimated from the preceding in-sample window and then applied to the subsequent out-of-sample holding period. The window was then rolled forward and the procedure repeated until the end of the sample. This design was preferred to a single in-sample optimisation because it reduced look-ahead bias and enabled repeated testing across different market conditions. In methodological terms, this rolling structure was also consistent with the evaluation patterns identified in the reviewed portfolio-optimisation literature.

A separate pilot study was not required because the research did not involve human participants, questionnaires, or competing candidate systems that first needed to be screened before main testing. Instead, preliminary implementation checks were carried out to verify price alignment, return calculation, optimisation constraints, and output consistency before running the full backtest. This was sufficient for establishing a stable quantitative workflow without introducing an unnecessary additional experimental phase.

E. Performance Evaluation

The realised out-of-sample portfolio results were evaluated using both return-oriented and downside-risk metrics. Return and risk-adjusted performance were assessed using CAGR, annualised volatility, Sharpe ratio, and Sortino ratio.

Downside-risk behaviour was assessed using VaR, CVaR, and maximum drawdown. These measures were selected because the hypothesis did not claim that CVaR-minimisation would maximise return; rather, it tested whether downside-risk control could be improved without materially worsening return-based performance.

The analysis was primarily comparative. The EW, MV, and CVaR portfolios were compared under the same dataset, constraints, and backtesting logic, and the resulting metrics were interpreted relative to the hypothesis and to the benchmark-comparison designs discussed in Section II. This approach was consistent with the assignment guidance that findings should later be discussed honestly and compared with existing research rather than being presented as a perfect or universal solution.

F. Validity, Reliability, Generalisability, Limitations, and Ethical Considerations

The validity of the study was strengthened by the controlled design. All three portfolio rules were tested on the same ETF basket, over the same time horizon, using the same adjusted-price treatment, date alignment, and investment constraints. The main intentional source of variation was therefore the portfolio-construction rule itself. Reliability was supported through the use of fixed procedures for data preparation, return calculation, portfolio construction, and backtesting, together with local storage of the downloaded data for reproducibility. However, the generalisability of the findings remained bounded by the specific six-ETF UCITS basket, the 2016–2025 period, and the long-only fully invested framework used in the study.

The study also had several limitations. First, it relied on a relatively small and intentionally transparent ETF universe rather than a large institutional asset set. Second, portfolio outcomes depended on historical market behaviour and therefore could not guarantee future performance. Third, the backtest was based on market-price data and did not attempt to model every real-world friction in depth, such as taxes or all trading frictions. These limitations do not invalidate the study, but they do define the scope within which its conclusions should be interpreted.

Finally, the study raised minimal ethical risk because it used secondary historical financial data and did not involve human participants, interviews, surveys, or personal data. The main ethical obligations were therefore academic honesty, correct citation, transparent reporting of the method, and accurate acknowledgment of the data source and implementation choices. This is consistent with the wider MCAST guidance that methodology should be rigorous, clearly explained, and properly referenced.

IV. FINDINGS & DISCUSSION OF RESULTS

A. Overview

Section IV presents the out-of-sample results for the three portfolio strategies examined in this study: Equal-Weight,

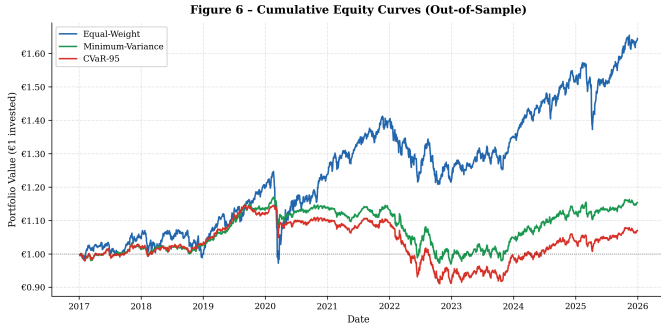


Fig. 4. Cumulative equity curves for Equal-Weight, Minimum-Variance, and CVaR-95 portfolios over the realised out-of-sample evaluation period.

Minimum-Variance, and CVaR-minimisation at the 95% confidence level. Results are evaluated using the seven metrics defined in Section III as the dependent variables for testing the main hypothesis, namely CAGR, annualised volatility, Sharpe ratio, Sortino ratio, VaR, CVaR, and maximum drawdown. Section IV.B reports the headline return and risk-adjusted performance of each strategy. Section IV.C then examines the downside-risk metrics directly relevant to the hypothesis. Section IV.D analyses the weight allocation patterns that underpin the observed performance outcomes. Section IV.E presents three sensitivity experiments to assess the robustness of the baseline findings. Section IV.F synthesises the results in relation to the main hypothesis, the reviewed literature, and the validity, reliability, and generalisability of the study.

B. Return and Risk-Adjusted Metrics

Table II summarises the out-of-sample performance of the three portfolio strategies over the 2017–2025 realised evaluation period. Although the full dataset covers 2016–2025, the first 252 trading days were used as the initial estimation window, meaning that reported portfolio outcomes begin after this first in-sample window. The table therefore reports only realised out-of-sample results.

The main return-based result is that Equal-Weight produced the strongest performance among the three strategies. It achieved the highest CAGR at 5.74%, compared with 1.61% for Minimum-Variance and 0.76% for CVaR-95. It also recorded the highest Sharpe ratio of 0.555 and the highest Sortino ratio of 0.741. This indicates that Equal-Weight did not only deliver the highest return, but also the strongest conventional risk-adjusted performance in this sample. This result is also visible in Fig. 4, where the Equal-Weight cumulative equity curve finishes above the two optimised portfolios. The supporting metrics comparison chart in Appendix A further confirms that Equal-Weight leads on the main return and risk-adjusted metrics.

However, this stronger return performance came with higher total volatility. Equal-Weight recorded annualised volatility of 10.06%, almost double that of Minimum-Variance at 5.59% and CVaR-95 at 5.46%. This shows that the Equal-Weight

portfolio accepted greater variation in returns in order to maintain broad exposure to all six ETFs. Since four of the six ETFs represented equity markets, the Equal-Weight strategy preserved a higher equity allocation throughout the out-of-sample period. This allowed it to benefit more from market upside than the two defensive strategies.

The lower return of Minimum-Variance and CVaR-95 is consistent with their optimisation objectives. These strategies were not designed to maximise return; they were designed to reduce variance or tail-risk exposure. Their lower volatility values show that the optimisers were successful in reducing overall return fluctuation. However, their much lower CAGR values show that this risk reduction came at a meaningful opportunity cost. In this sample, the reduction in risk did not translate into stronger risk-adjusted performance because the defensive allocation also reduced participation in equity-market growth.

This result is important for interpreting the hypothesis. The hypothesis did not claim that CVaR-minimisation would maximise return. Instead, it proposed that CVaR-minimisation would improve downside-risk control without material deterioration in CAGR or Sharpe ratio. The return and risk-adjusted metrics therefore already show a potential weakness in the hypothesis: CVaR-95 achieved lower volatility than Equal-Weight, but it also produced a much lower CAGR and Sharpe ratio. The next subsection therefore examines whether the downside-risk metrics are strong enough to justify this return sacrifice.

C. Downside-Risk Comparison

The downside-risk results provide a more direct test of the main hypothesis because the study is centred on CVaR-based risk control. In Table II, CVaR-95 achieved the lowest VaR and CVaR values among the three strategies. Its VaR at the 95% level was 0.54%, slightly lower than Minimum-Variance at 0.55% and substantially lower than Equal-Weight at 0.95%. Its CVaR was also the lowest at 0.82%, compared with 0.84% for Minimum-Variance and 1.56% for Equal-Weight. Since VaR and CVaR are reported as positive loss magnitudes, lower values indicate better short-horizon tail-risk control.

This finding supports one part of the hypothesis. The CVaR-minimisation objective was effective in reducing the lower tail of the daily return distribution. The improvement over Minimum-Variance was small, but it was still in the expected direction. The improvement over Equal-Weight was larger, which is consistent with the fact that Equal-Weight maintained higher equity exposure and therefore accepted greater downside variation. In this sense, the CVaR strategy behaved as expected: it reduced the short-horizon tail-loss metrics that it was designed to control.

However, the downside-risk result was not uniformly favourable to CVaR-95. Maximum drawdown shows a different pattern. Minimum-Variance achieved the shallowest maximum drawdown at -17.13% , while CVaR-95 recorded -20.55% and Equal-Weight recorded -22.02% . This means that CVaR-95 did not achieve the lowest peak-to-trough loss

TABLE II
OUT-OF-SAMPLE PERFORMANCE SUMMARY.

Strategy	CAGR	Ann. Volatility	Sharpe	Sortino	VaR 95%	CVaR 95%	Max Drawdown
Equal-Weight	5.74%	10.06%	0.555	0.741	0.95%	1.56%	-22.02%
Minimum-Variance	1.61%	5.59%	0.286	0.392	0.55%	0.84%	-17.13%
CVaR-95	0.76%	5.46%	0.138	0.191	0.54%	0.82%	-20.55%

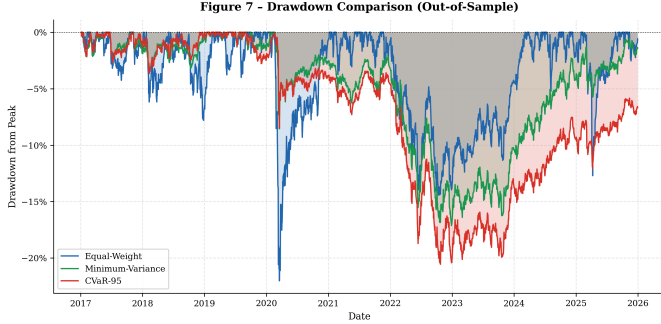


Fig. 5. Drawdown comparison for Equal-Weight, Minimum-Variance, and CVaR-95 portfolios.

over the evaluation period. Fig. 5 also supports this distinction, as the CVaR-95 portfolio generally experienced deeper and more persistent drawdowns than the Minimum-Variance portfolio, especially from 2020 onward.

This difference is important because VaR/CVaR and maximum drawdown measure different aspects of risk. VaR and CVaR summarise short-horizon tail losses in the return distribution. Maximum drawdown instead measures the worst cumulative decline from a previous peak. A portfolio can therefore reduce daily tail losses while still experiencing a weaker cumulative recovery path. The results show exactly this pattern: CVaR-95 performed best on VaR and CVaR, but Minimum-Variance performed best on maximum drawdown.

Therefore, the downside-risk evidence only partially supports the hypothesis. CVaR-95 achieved the lowest VaR and CVaR, which supports the claim that CVaR-minimisation improves short-horizon tail-risk control. However, it did not achieve the lowest maximum drawdown. As a result, the CVaR strategy cannot be described as the strongest downside-risk strategy across all risk measures. It is more accurate to state that CVaR-95 improved the specific daily tail-risk metrics it targeted, but did not dominate the Minimum-Variance benchmark on longer path-dependent downside risk.

D. Weight Allocation Analysis

The allocation evidence helps explain why the three strategies produced different return and risk outcomes. The detailed weight evolution plots are provided in Appendix A. Appendix A, Fig. 7 shows that Equal-Weight maintained a constant allocation of approximately 16.67% to each ETF. This gave the strategy a stable combined equity exposure of around two-thirds of the portfolio, since four of the six ETFs represented equity markets. This structure explains why Equal-Weight

captured more market upside and achieved the highest CAGR, but also why it carried the highest annualised volatility and higher tail-risk values.

By contrast, Appendix A, Figs. 8 and 9 show that the optimised strategies allocated much larger weights to the two bond ETFs, especially SXRQ and, at several points, CBU0. This pattern was particularly visible in the Minimum-Variance and CVaR-95 portfolios. The allocation behaviour is consistent with the objectives of the two optimisers. Minimum-Variance reduced exposure to assets that increased portfolio variance, while CVaR-95 reduced exposure to assets that contributed more strongly to historical tail losses.

Appendix A, Fig. 10 provides further support for this behaviour through the ETF correlation heatmap. The equity ETFs were positively correlated with each other, while the bond ETFs showed lower or near-zero correlations with most equity ETFs. This made the bond ETFs useful for reducing portfolio risk. As a result, the optimised strategies shifted towards defensive assets that reduced volatility and short-horizon tail risk. This helps explain why Minimum-Variance and CVaR-95 recorded much lower annualised volatility than Equal-Weight.

The same allocation behaviour also explains the weaker return performance of the optimised portfolios. By concentrating more heavily in bond ETFs, Minimum-Variance and CVaR-95 reduced their exposure to equity-market growth. This was beneficial for volatility reduction, but it limited participation in the upside that drove the Equal-Weight result. The findings therefore show a clear allocation-based trade-off: the optimisers reduced risk by moving towards defensive assets, but this also reduced their ability to generate return during the out-of-sample period.

This allocation analysis answers the portfolio-construction research question more directly. Under identical long-only and fully invested constraints, CVaR-minimisation behaved differently from Equal-Weight and broadly similarly to Minimum-Variance. It did not create a balanced return-seeking portfolio; it created a defensive portfolio. This behaviour is consistent with a downside-risk objective, but it also explains why the strategy struggled to remain competitive on CAGR, Sharpe ratio and Sortino ratio.

E. Sensitivity and Robustness

Three sensitivity experiments were conducted to examine whether the baseline conclusions depended strongly on the selected backtest parameters. The tests varied the estimation window length, rebalancing frequency, and CVaR confidence level. These checks were important because the baseline

specification used a 252-day estimation window, a 21-trading-day rebalance frequency, and a CVaR confidence level of 0.95. Robustness analysis therefore helped assess whether the findings were stable or mainly driven by one parameter setting.

The estimation-window test compared the 252-day baseline with a longer 504-day window. The longer window did not improve the main results. For CVaR-95, the Sharpe ratio declined from 0.138 under the 252-day window to 0.110 under the 504-day window. Minimum-Variance was affected more strongly, with its Sharpe ratio falling from 0.286 to 0.084. The 504-day setting also reduced the number of out-of-sample observations from 2246 to 1994, removing approximately one trading year from the realised evaluation period. This means that the longer estimation window did not provide enough benefit to justify the shorter out-of-sample period. The 252-day window was therefore retained as the more defensible baseline because it provided a full year of estimation data while preserving a longer evaluation period.

The rebalancing-frequency test compared the monthly baseline of 21 trading days with a quarterly alternative of 63 trading days. Equal-Weight produced identical results under both settings, with a Sharpe ratio of 0.555, a CAGR of 5.74%, and a maximum drawdown of -22.02% . This is expected because Equal-Weight uses the same constant $1/N$ target allocation and the study did not model transaction costs. CVaR-95 was also relatively stable: its Sharpe ratio changed only slightly from 0.138 to 0.130, and maximum drawdown remained close to -20.6% . Minimum-Variance improved under quarterly rebalancing, with its Sharpe ratio increasing from 0.286 to 0.322 and its maximum drawdown improving from -17.13% to -14.94% . This suggests that less frequent rebalancing may have reduced sensitivity to short-term covariance changes in this sample. However, the overall conclusion did not change, since Equal-Weight remained strongest on return and Sharpe performance.

The CVaR confidence-level test compared alpha values of 0.90, 0.95 and 0.99. CVaR-90 and CVaR-95 produced broadly similar results. CVaR-90 achieved a CAGR of 0.79%, a Sharpe ratio of 0.147, a CVaR of 0.82%, and a maximum drawdown of -20.46% . These were close to the CVaR-95 baseline, which achieved a CAGR of 0.76%, a Sharpe ratio of 0.138, a CVaR of 0.82%, and a maximum drawdown of -20.55% . This suggests that the baseline alpha value was not highly sensitive relative to the 90% alternative.

By contrast, the 99% confidence level was counterproductive. CVaR-99 recorded a lower CAGR of 0.23%, a lower Sharpe ratio of 0.039, a higher annualised volatility of 5.93%, a higher realised CVaR of 0.89%, and a deeper maximum drawdown of -22.56% . One explanation is that, with a 252-day estimation window, the 1% tail contains only about 2.5 observations. The optimisation is therefore driven by a very small number of extreme observations in each rolling window, which can make the resulting allocation less stable out of sample. For this reason, $\alpha = 0.95$ was retained as the most defensible baseline setting.

Overall, the sensitivity analysis strengthens rather than

weakens the baseline interpretation. CVaR-95 remained a defensive strategy under alternative window and rebalance settings, but it did not become the best return-adjusted performer. The 99% confidence level also showed that making the CVaR objective more extreme did not improve realised tail-risk control in this dataset. The robustness evidence therefore supports the main conclusion that CVaR-minimisation reduced daily tail-risk metrics but involved a material return and Sharpe-ratio trade-off. Supporting sensitivity plots for the estimation-window, rebalancing-frequency, and CVaR-confidence-level experiments are provided in Appendix A, Figs. 11–13.

F. Discussion

The findings provide mixed evidence for the main hypothesis. The CVaR-95 strategy achieved the lowest CVaR value at 0.82%, compared with 0.84% for Minimum-Variance and 1.56% for Equal-Weight. This supports the part of the hypothesis stating that CVaR-minimisation should improve tail-risk control. However, the same strategy did not achieve the lowest maximum drawdown. Minimum-Variance recorded the best maximum drawdown at -17.1% , compared with -20.6% for CVaR-95. In addition, CVaR-95 suffered a material deterioration in return-adjusted performance, with a Sharpe ratio of 0.138 compared with 0.555 for Equal-Weight. The hypothesis is therefore partially supported rather than fully supported.

The most reasonable interpretation is that CVaR-minimisation acted as a defensive allocation rule, not as a return-maximising strategy. Its lower VaR and CVaR values show that it was effective for short-horizon tail-loss control. However, the weight evolution analysis shows that this protection was achieved by concentrating more heavily in government bond ETFs. That allocation reduced volatility and tail-risk exposure, but it also reduced participation in equity-market growth. This explains why CVaR-95 had lower annualised volatility than Equal-Weight but much lower CAGR and Sharpe performance.

This result is consistent with the wider portfolio-optimisation literature discussed in Section II. Muñoz and Cifuentes [1] show that the choice of risk metric can materially affect portfolio allocations and risk profiles, but does not necessarily guarantee simple dominance by one formulation. The present study supports that point in a smaller and more transparent UCITS ETF setting. CVaR changed the allocation and improved the specific tail-risk measure it targeted, but it did not produce superior overall performance. Similarly, Teplova et al. [3] evaluate CVaR-based ETF portfolios using benchmark comparisons and downside-risk measures, but their framework includes additional modelling elements such as Black-Litterman views and copula-based dependence modelling. The present study shows that a simpler CVaR-minimisation design is easier to interpret, but may also be more limited in its ability to improve return-adjusted performance.

The findings also show why the Equal-Weight benchmark is important. Equal-Weight is simple, but it remained difficult to beat in this sample. Its constant exposure to all six ETFs

allowed it to benefit from equity-market appreciation during the out-of-sample period. This does not invalidate the CVaR result, because the CVaR strategy did reduce VaR and CVaR. However, it shows that downside-risk optimisation can sacrifice too much upside when the market environment rewards equity exposure. The result therefore supports a balanced interpretation: CVaR was useful for daily tail-risk control, but Equal-Weight was superior for return and conventional risk-adjusted performance.

From a validity perspective, the findings are supported by the controlled experimental design. All three strategies were tested on the same ETF universe, using the same EUR-aligned return data, long-only fully invested constraints, rolling backtest logic, and evaluation metrics. This strengthens internal validity because the main intended difference between the strategies was the portfolio-construction rule. Reliability was also supported by the repeatable data preparation, backtesting, metric calculation, and sensitivity-analysis pipeline.

However, the findings should not be generalised beyond the scope of the study. The ETF universe contained only six UCITS ETFs, and the realised out-of-sample period was 2017–2025. Different ETF baskets, different asset classes, different fee assumptions, transaction-cost modelling, or a different market period could produce different results. The study should therefore be read as a controlled benchmark comparison in a Malta-accessible UCITS ETF context, not as universal proof that CVaR-minimisation is either superior or inferior in all portfolio settings.

Overall, the results answer the main research question by showing that CVaR-minimisation can improve daily tail-risk metrics compared with Equal-Weight and Minimum-Variance, but not without a material deterioration in return and Sharpe ratio. The practical implication is that CVaR-minimisation may be suitable where short-horizon tail-loss control is the main priority. However, for investors who also require competitive return and risk-adjusted performance, a simple CVaR-minimisation rule may need to be combined with additional constraints, return targets, or broader optimisation design.

V. CONCLUSION

This study examined whether CVaR-minimisation provides stronger downside-risk control than Equal-Weight and Minimum-Variance portfolio construction in a Malta-accessible UCITS ETF basket. The research was structured around three questions covering the dataset, the portfolio-construction process, and the evaluation of out-of-sample performance.

RQ1 concerned the dataset and investable basket. The study showed that a fixed six-ETF UCITS basket could be used to create a consistent and transparent dataset for portfolio backtesting. The selected ETFs provided exposure to major equity markets and government-bond sleeves, while EUR alignment allowed the strategies to be compared under a common reporting currency. This provided a suitable controlled dataset for the scope of the research, although it remained smaller than

the broader asset universes used in some portfolio-optimisation studies.

RQ2 concerned the portfolio-construction process. Under identical long-only and fully invested constraints, CVaR-minimisation behaved as a defensive allocation rule. It did not behave like Equal-Weight, which maintained constant exposure to all six ETFs, but was closer to Minimum-Variance in its preference for lower-risk and lower-correlation bond exposures. This confirmed that changing the portfolio-construction rule affected the resulting allocation, which is consistent with the reviewed literature on risk-metric choice and portfolio behaviour [1], [3].

RQ3 concerned evaluation and testing. The out-of-sample results showed that Equal-Weight achieved the strongest return and risk-adjusted performance, with the highest CAGR and Sharpe ratio. CVaR-95 achieved the lowest VaR and CVaR values, which supports the part of the hypothesis related to daily tail-loss control. However, it did not achieve the lowest maximum drawdown, and its CAGR and Sharpe ratio were materially lower than Equal-Weight. Minimum-Variance achieved the best maximum drawdown. Therefore, the main hypothesis was only partially supported: CVaR-minimisation improved the specific tail-risk metrics it targeted, but it did not dominate the benchmark strategies across all downside-risk and return-adjusted measures.

Overall, the study achieved its main aim to a reasonable extent. It implemented the three strategies under the same dataset, constraints, rolling backtest design, and evaluation metrics, and produced a clear empirical comparison. However, the results also showed that a simple CVaR-minimisation framework is not automatically superior. Its defensive allocation reduced short-horizon tail-risk, but this came at the cost of weaker participation in equity-market growth. This explains why the CVaR strategy reduced VaR and CVaR, but did not preserve competitive CAGR or Sharpe performance.

The findings should be interpreted within the limits of the research design. The study used a small six-ETF UCITS basket, a specific 2017–2025 realised out-of-sample period, market-price data, and a simplified backtest without a full transaction-cost or tax model. These choices supported transparency and feasibility, but they also limit generalisability. The results should therefore be read as evidence from a controlled Malta-accessible ETF comparison, not as a universal conclusion about all CVaR-based portfolios.

Future research should extend this work in several directions. First, a larger UCITS ETF universe could be tested to see whether CVaR-minimisation performs differently when more asset classes and regions are available. Second, future work should include transaction costs, bid-ask spreads, taxes, and fund fees to make the backtest closer to real investor implementation. Third, researchers could compare exchange market-price data with official NAV-based fund data where available. Fourth, more advanced CVaR frameworks, such as mean-CVaR, target-return CVaR, Black-Litterman mean-CVaR, or regime-based optimisation, could be tested against the simple CVaR-minimisation approach used here. Finally,

future studies could examine whether CVaR performs better during specific stress periods rather than over the whole out-of-sample period. These extensions would help determine whether the weakness found in this study was due to the CVaR objective itself, the small ETF universe, or the chosen market period.

APPENDIX A SUPPORTING FIGURES

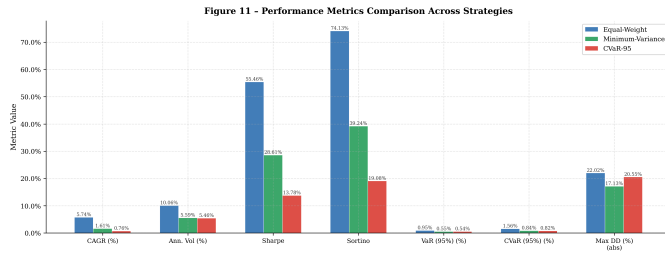


Fig. 6. Comparison of performance and downside-risk metrics across the three strategies.

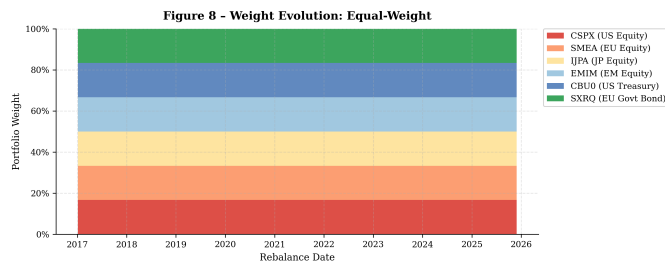


Fig. 7. Equal-Weight allocation through time.

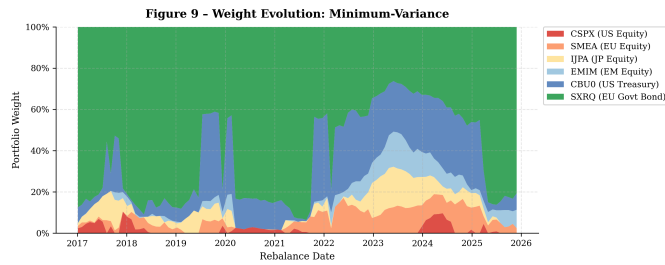


Fig. 8. Minimum-Variance allocation through time.

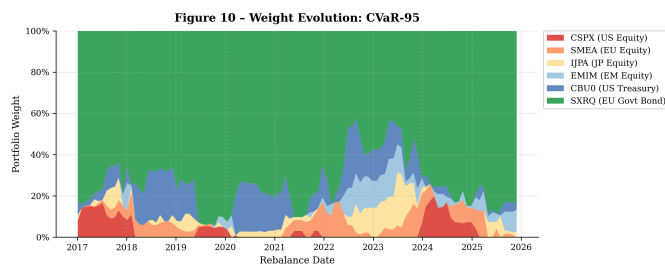


Fig. 9. CVaR-95 allocation through time.

Figure 12 - ETF Daily Log-Return Correlation Matrix (2016-2025)

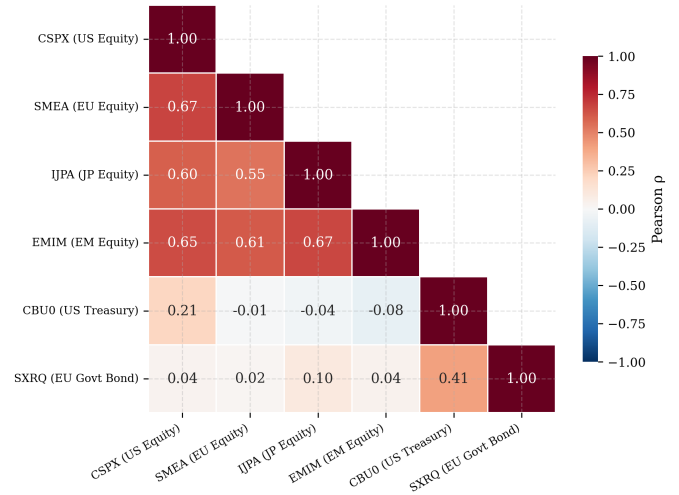


Fig. 10. Pearson correlation heatmap for the ETF return series.

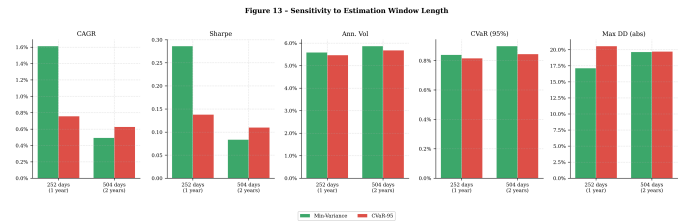


Fig. 11. Sensitivity of strategy performance to estimation-window length.

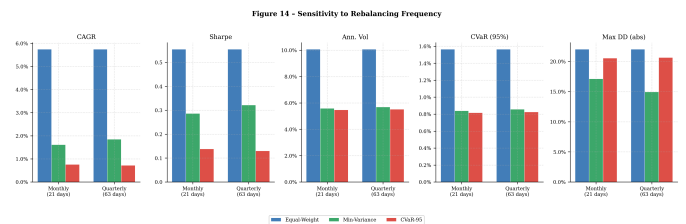


Fig. 12. Sensitivity of strategy performance to rebalancing frequency.

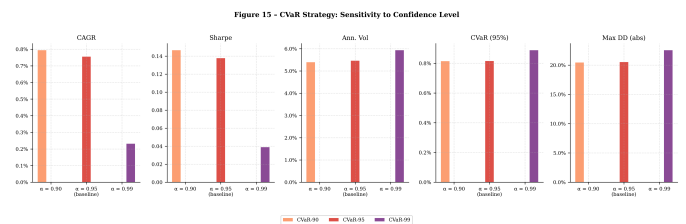


Fig. 13. Sensitivity of CVaR strategy performance to confidence level.

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